

VSTACK + CHOOSECOLS + FILTER in Excel

1. What is VSTACK + CHOOSECOLS + FILTER?

These three functions work together. FILTER keeps only the rows that meet a condition, CHOOSECOLS keeps only the columns you need from those rows, and VSTACK stacks the results from several tables into one list. Together they collect matching records from different tables, such as different regions or months. A Data Analyst uses them to build one clean report from several sources.

2. Syntax

`=VSTACK(CHOOSECOLS(FILTER(array1, include1), col_num1, [col_num2], ...), CHOOSECOLS(FILTER(array2, include2), col_num1, [col_num2], ...))`

Function	Argument	Required?	What it accepts
FILTER	array	Required	The range to filter (for example A2:D4)
FILTER	include	Required	A TRUE/FALSE test for each row, such as B2:B4="Laptop". Join tests with * (AND) or + (OR).
CHOOSECOLS	col_num1, ...	col_num1 required	The columns to keep from the filtered rows, counted inside the array. More numbers are optional.
VSTACK	array1, ...	array1 required	The CHOOSECOLS results to stack. More arrays are optional.

Excel works from the inside out: FILTER first, then CHOOSECOLS, then VSTACK. Every CHOOSECOLS part must return the same columns in the same order.

3. Simple Example

Collect the laptop sales from the North and East tables:

	A	B	C	D	E	F	G
1	ID	Product	Region	Sales		Product	Sales
2	P001	Laptop	North	1200		Laptop	1200
3	P002	Mouse	North	150		Laptop	800
4	ID	Product	Region	Sales			
5	P004	Laptop	East	800			
6	P005	Mouse	East	250			

Formula in F2 only (headings in F1:G1 are typed):
`=VSTACK(CHOOSECOLS(FILTER(A2:D3, B2:B3="Laptop"), 2, 4), CHOOSECOLS(FILTER(A5:D6, B5:B6="Laptop"), 2, 4))`
FILTER keeps the Laptop row in each table. CHOOSECOLS keeps columns 2 (Product) and 4 (Sales). VSTACK stacks the two results.
Result: Laptop 1200 and Laptop 800

4. Another Example

Collect North orders over 300 from the Jan and Feb tables:

	A	B	C	D	E	F
1	Order	Region	Amount		Order	Amount
2	O101	North	450		O101	450
3	O102	South	600		O201	520
4	Order	Region	Amount		O202	310
5	O201	North	520			
6	O202	North	310			

Formula in E2 only (headings in E1:F1 are typed):
`=VSTACK(CHOOSECOLS(FILTER(A2:C3, (B2:B3="North")*(C2:C3>300)), 1, 3), CHOOSECOLS(FILTER(A5:C6, (B5:B6="North")*(C5:C6>300)), 1, 3))`
The * means both conditions must be true. Jan gives O101. Feb gives O201 and O202. **Result: 3 rows**

5. When to Use It

- Collect matching records from several tables, such as one table per region or month
- Compare one product or customer across different tables
- Combine high-value orders from several sources into one list
- Build report lists that show only the columns you need
- Create dynamic lists for dashboards that update when the data changes

6. Important Things to Remember

- Read the formula from the inside out: FILTER, then CHOOSECOLS, then VSTACK.
- Each CHOOSECOLS part must return the same columns in the same order. Fewer columns give #N/A.
- If FILTER finds no match in one table, it returns #CALC! and the whole formula shows an error. Wrap it in IFERROR to show a message.
- Column numbers in CHOOSECOLS count inside the filtered array. Text in conditions goes in quotes, and each condition goes in brackets.
- The result spills into neighboring cells. They must be empty, or Excel shows #SPILL!.
- CHOOSECOLS and VSTACK need Microsoft 365 or Excel 2024. This formula does not work in Excel 2021, even though FILTER does.

7. Common Mistakes

Mistake: One table has no matching rows, so the whole formula shows #CALC!. **Fix:** Wrap it in IFERROR, such as `=IFERROR(VSTACK(...), "No matches")`. The `if_empty` argument does not help, because CHOOSECOLS then gives #VALUE!.
Mistake: Choosing different columns in each CHOOSECOLS, so #N/A appears or values land under the wrong headings. **Fix:** Use the same column numbers in every part.
Mistake: Forgetting the brackets around each condition, such as `B2:B4="North"*C2:C4>300`. **Fix:** Write `(B2:B4="North")*(C2:C4>300)`.

8. Data Analyst Tip

If your data is in Excel Tables, use table names such as `Table1[Region]` in the formula, so it picks up new rows automatically as the tables grow.